

OTShield: key features and functionality

One integrated platform, two cooperating modes sharing a common data layer. See the key at the end for the Status column.

Feature	What it does (function)	Customer need / challenge it meets	Status
SOC Mode - deception-based active defence			
Protocol-aware decoy fabric	Realistic decoy HMIs/controllers (Siemens WinCC, Rockwell, Schneider, generic) speaking Modbus, S7Comm, EtherNet/IP, OPC UA, DNP3, IEC 104.	Proactive CNI defence: any decoy contact is hostile, giving high-confidence, low-noise detection.	Live today
Purdue-aware deep-packet inspection	Deep-packet inspection and protocol dissectors across Purdue zones; east-west and north-south traffic analysis.	Detects sophisticated actors moving inside OT without touching live controllers.	Live today
AI/ML anomaly detection on OT traffic	New ML models detect deviations from normal OT behaviour and unexpected protocol commands; trained and validated on the operator's real traffic and live decoy data.	Catches novel and stealthy intrusions that static rules miss, with low false positives.	Project (new R&D)
AI attacker-behaviour classification	ML classifies attacker TTPs from decoy and PCAP data, clusters campaigns and auto-maps them to MITRE ATT&CK for ICS.	Faster detection and eviction of sophisticated actors; less analyst load.	Project (new R&D)
Asset inventory + risk scoring	Auto-discovers PLCs, RTUs, HMIs, engineering workstations; Purdue mapping and risk score.	Surfaces legacy-system exposure in CNI.	Live today
MITRE ATT&CK for ICS mapping	Maps attacker TTPs across the full ICS kill chain; IOC export to SIEM / SOAR / TAXII / MISP.	Supports detection, eviction and incident response.	Live today
Energy-specific decoys & dissectors	Decoys and dissectors tuned to the operator's real energy assets and protocols.	Fidelity for the host energy environment.	Project
Research Mode - offline analyst copilot			
Research workbench	Library, Summary, Inventory, Ports & Services, Threads, Findings, Vulns; runs fully offline.	Makes analysts effective without sending data off the plant.	Live today
Offline LLM-assisted threat analysis	Local LLM copilot summarises and prioritises detections, answers with cited sources and turns activity into auditable Findings; interprets the AI detection-layer output for analysts.	Practical, safe threat hunting and faster triage for scarce CNI analysts.	Live today / Project
Threat hunting over shared data	Analysts query detections, manuals and asset data together and follow leads safely.	Threat hunting for CNI organisations with zero risk to live plant.	Project
Calibrated confidence + audit	Confidence labelling and an auditable evidence ledger for each finding.	Defensible results for security review and regulators.	Project

Shared platform, deployment and assurance

Feature	What it does (function)	Customer need / challenge it meets	Status
Shared platform capabilities (both modes)			
Shared data layer	SOC and Research modes share one data layer; live attacker behaviour informs research and research informs defence.	One integrated solution rather than separate tools.	Live today
Air-gap-safe threat-intel feed	Live internet-exposed decoy fabric captures real ICS attacker tactics, packaged as signed update bundles ingested offline.	Keeps an air-gapped plant current without internet.	Live today / Project
Coexistence with incumbents	Coexists with Dragos / Claroty / Nozomi; adds the deception and copilot layers they lack.	Procurement-friendly: no rip-and-replace.	Live today
Deployment, integration and assurance			
Flexible deployment	Cisco-compatible hardware appliance or software-only; deploy in weeks.	Fits the customer's existing stack; short sales cycle.	Live today
SIEM / SOAR integration	Bidirectional alert and IOC export; playbook-ready alert format.	Plugs into the operator's existing SOC tooling.	Live today / Project
Standards alignment	Aligned with NIS / NIS2, IEC 62443 and the NCSC CAF.	Meets the frameworks operators are audited against.	Live today
TRL 8 evidence + SBOM	Adjudicated benchmark report and signed CycloneDX software bill of materials.	Operator-accepted, qualified, auditable supply chain.	Project
Open-source core	Built on Apache 2.0 / MIT components; no closed-model or licence lock-in.	Lower cost and clear freedom to operate.	Live today

Key: **Live today** = operational now (TRL 7). **Project** = delivered or qualified during the 12-month project (to TRL 8).